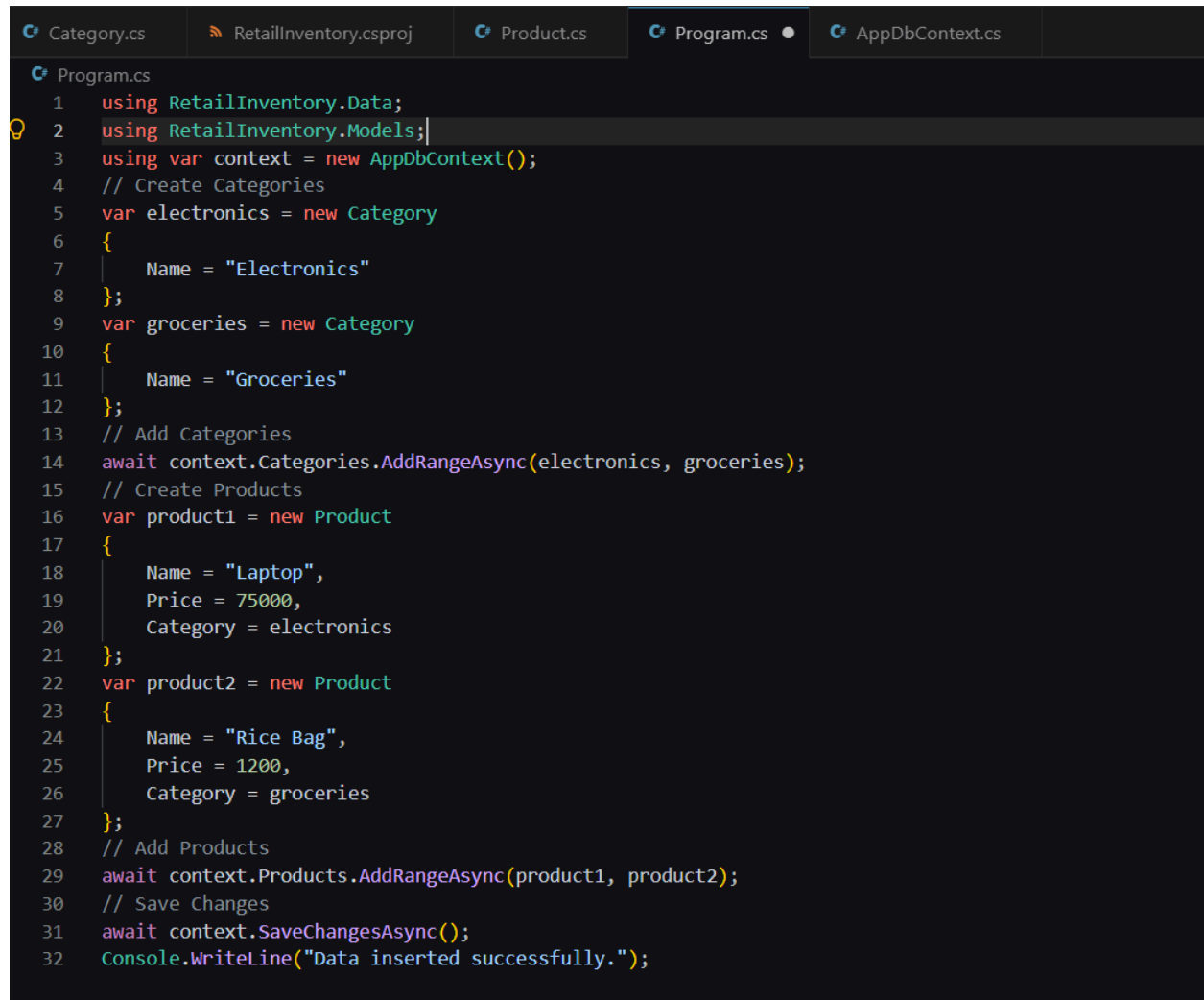


Lab 4: Inserting Initial Data into the Database

Aim

To insert initial data into the SQL Server database using Entity Framework Core's `AddRangeAsync()` and `SaveChangesAsync()` methods.

Insert data in [program.cs](#)



```
1 using RetailInventory.Data;
2 using RetailInventory.Models;
3 using var context = new AppDbContext();
4 // Create Categories
5 var electronics = new Category
6 {
7     Name = "Electronics"
8 };
9 var groceries = new Category
10 {
11     Name = "Groceries"
12 };
13 // Add Categories
14 await context.Categories.AddRangeAsync(electronics, groceries);
15 // Create Products
16 var product1 = new Product
17 {
18     Name = "Laptop",
19     Price = 75000,
20     Category = electronics
21 };
22 var product2 = new Product
23 {
24     Name = "Rice Bag",
25     Price = 1200,
26     Category = groceries
27 };
28 // Add Products
29 await context.Products.AddRangeAsync(product1, product2);
30 // Save Changes
31 await context.SaveChangesAsync();
32 Console.WriteLine("Data inserted successfully.");
```

Run the App:

```

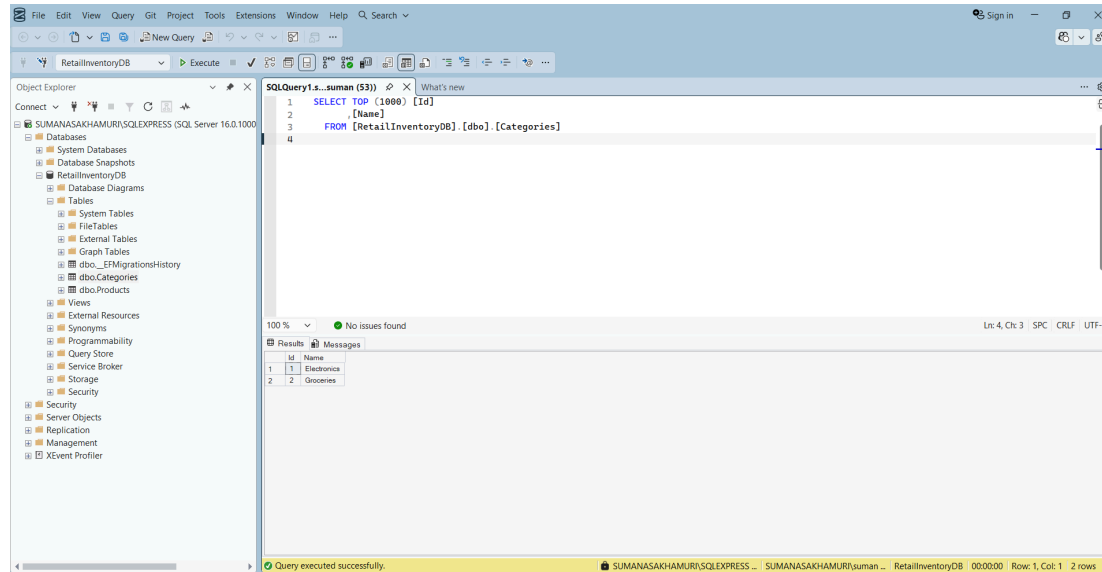
PS C:\Users\suman\OneDrive\Desktop\RetailInventory> dotnet build
Restore complete (0.9s)
  RetailInventory succeeded (6.7s) → bin\Debug\net9.0\RetailInventory.dll

Build succeeded in 9.0s
PS C:\Users\suman\OneDrive\Desktop\RetailInventory> dotnet run
Data inserted successfully.

```

Verify in SQL server:

Category:



The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'SUMANASAKHAMURI\SQLEXPRESS (SQL Server 16.0.1000)'. The central query window shows the following SQL query:

```

1 SELECT TOP (1000) [Id]
2   [Name]
3 FROM [RetailInventoryDB].[dbo].[Categories]
4

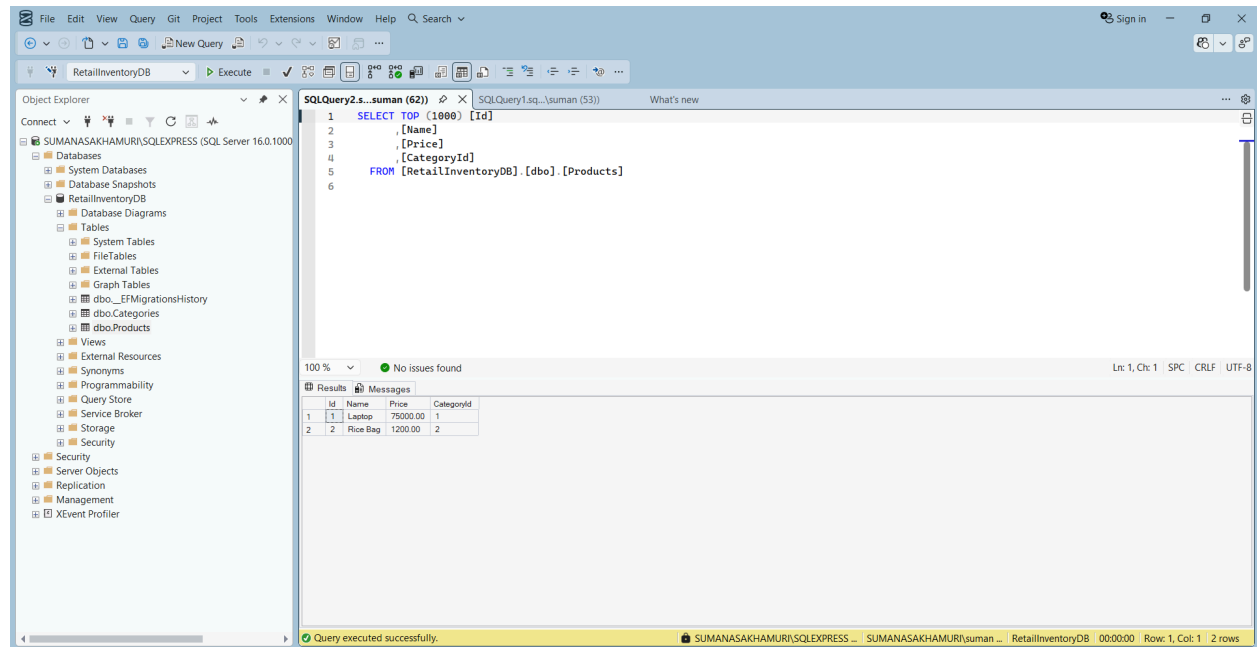
```

The Results pane at the bottom displays the following data:

Id	Name
1	Electronics
2	Groceries

The status bar at the bottom indicates 'Query executed successfully.' and 'SUMANASAKHAMURI\SQLEXPRESS ... SUMANASAKHAMURI\suman ... RetailInventoryDB 00:00:00 Row: 1, Col: 1 2 rows'.

Product:



The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'SUMANASAKHAMURI\SQLEXPRESS (SQL Server 16.0.1000)'. The central query window shows the following SQL query:

```

1 SELECT TOP (1000) [Id]
2   [Name]
3   [Price]
4   [CategoryId]
5 FROM [RetailInventoryDB].[dbo].[Products]
6

```

The Results pane at the bottom displays the following data:

Id	Name	Price	CategoryId
1	Laptop	75000.00	1
2	Rice Bag	1200.00	2

The status bar at the bottom indicates 'Query executed successfully.' and 'SUMANASAKHAMURI\SQLEXPRESS ... SUMANASAKHAMURI\suman ... RetailInventoryDB 00:00:00 Row: 1, Col: 1 2 rows'.